

CHAPTER V

VELOCITY OF LIGHT

365. Finite Velocity of Light—Römer.—An entirely new era in the history of the science of light was introduced by the Danish astronomer Römer in 1676, when he not only showed that light did not travel instantaneously, as had been previously supposed, but also measured the velocity with which light travels through interplanetary space.

The planet Jupiter has four moons, and as these revolve round the planet they disappear once in each revolution, for when they pass into the shadow of the planet cast by the sun they become invisible, and are said to be eclipsed, for we only see the planets and their satellites by the light of the sun which they reflect.

If any one of Jupiter's moons revolves round the planet with a uniform angular velocity, as is the case with our moon, then the time which elapses between one passage of the moon into the shadow and the next ought to be constant, since it would be equal to the periodic time of the moon's revolution round the planet. However, if the times between successive eclipses of the planet are noted, as seen from the earth, it is found that they are not all equal.

Römer accounted for this phenomenon by supposing that light took an appreciable time to travel from Jupiter to the earth. Let S (Fig. 342) be the sun, E the earth, and J Jupiter; and suppose that when an eclipse of one of the moons occurs these three are in the relative positions shown, the earth being at its nearest point to Jupiter. When the next eclipse occurs, the earth will have moved round in its orbit to some such position as E_1 , while Jupiter will have only moved

a short distance round in its orbit. Hence the distance between the earth and Jupiter is now greater than before. If V is the speed with which light travels, then the observed time at which the eclipse occurs when the earth is at E will be at a time $\overline{EJ}/V$, *i.e.* the time taken by the light to traverse the space $\overline{EJ}$, after the actual eclipse. When the

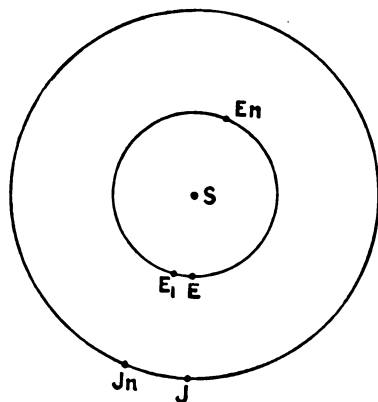


FIG. 342.

earth is at E_1 , the observed time will be $\overline{E_1J}/V$ later than the actual time. Hence, if θ is the actual time between two successive eclipses, the observed interval is $\theta + \overline{E_1J}/V - \overline{EJ}/V$, and since $\overline{E_1J}$ is greater than $\overline{EJ}$, the observed interval will be greater than the true interval; since, however, we cannot observe the *true* interval θ , the quantity V cannot be calculated from the observation of a single interval in this way.

At each successive eclipse the earth will be further and further away from Jupiter, till finally they come into the positions J_n, E_n , when they are at their maximum distance apart. After this, at each successive eclipse the distance will diminish, and hence the observed interval be less than the true interval.

Suppose that n eclipses occur between that which occurs when Jupiter and the earth are nearest together (at conjunction), and that which occurs when they are at their greatest distance (opposition), so that the actual interval between the first and last of these eclipses is $n\theta$. The observed interval is $n\theta + \overline{E_nJ_n}/V - \overline{EJ}/V$, or if d is the diameter of the earth's orbit, so that $\overline{E_nJ_n} - \overline{EJ} = d$, the observed interval T_1 is $n\theta + d/V$.

The actual interval between the eclipse when the earth is at opposition and the one when it is again at conjunction will also be $n\theta$. The observed interval, T_2 , since at the end of the series the earth is nearer Jupiter than at the commencement by a distance d , will be $n\theta - d/V$.

$$\text{Thus} \quad T_1 = n\theta + \frac{d}{V},$$

$$\text{and} \quad T_2 = n\theta - \frac{d}{V};$$

$$\therefore \quad T_1 - T_2 = \frac{2d}{V},$$

$$\text{or} \quad V = \frac{2d}{T_1 - T_2}.$$

Hence if we know the diameter, d , of the earth's orbit, we can calculate the velocity of light, V , from the difference between the interval T_1 which elapses between an eclipse at conjunction and the eclipse at the next opposition, and the interval T_2 between this eclipse at opposition and the one which occurs at the next conjunction.

The moon chosen was the innermost, which makes a revolution in about $1\frac{1}{2}$ days, and it was found that, starting at conjunction, the interval between the first eclipse of this planet and the 113th (when the earth and Jupiter came into opposition) exceeded the interval between the 113th and the 225th (when the earth and Jupiter were again in conjunction) by 33.2 minutes, and hence $T_1 - T_2 = 33.2$ minutes. If d , or the diameter of the earth's orbit, is taken as 195,600,000 miles, or 298,600,000 kilometres, this gives 186,300 miles per second, or 299,800 kilometres per second, as the velocity of light.

366. Fizeau's Method of Measuring the Velocity of Light.—The accuracy of the determination of the velocity of light by Rømer's method is limited by the accuracy with which we know the diameter of the earth's orbit, hence it is important to determine the velocity of light between two terrestrial points, the distance between which can be directly

measured. The first to perform this experiment was Fizeau, who in 1849 measured the velocity of light by using a method depending on the eclipsing of a source of light by the teeth of a rapidly rotating wheel, the principle of the experiment resembling Römer's method.

A bright source of light was placed at *L* (Fig. 343), and after passing through a lens *A*, a certain proportion of the rays of light was reflected from the surface of an unsilvered plate of glass, *G*, placed at an angle of 45° . The reflected rays came to a focus at *F*, this point being the principal focus of a second lens *B*. Thus the light left *B* in a parallel beam, which, after traversing a distance of about four miles, fell on a lens *C*, and was brought to a focus at the surface of a spherical mirror *D*. The curvature of this mirror is such that the lens *C* is at its centre of curvature, and hence the rays are reflected back along their path, so that on emerging from the lens *C* they again form a parallel beam. This reflected beam falls on the lens *B*, is brought to a focus at *F*, and then falls on the plate of glass *G*, where some of the rays will be reflected, and some will be transmitted and enter the eye-piece *E*, so that a bright

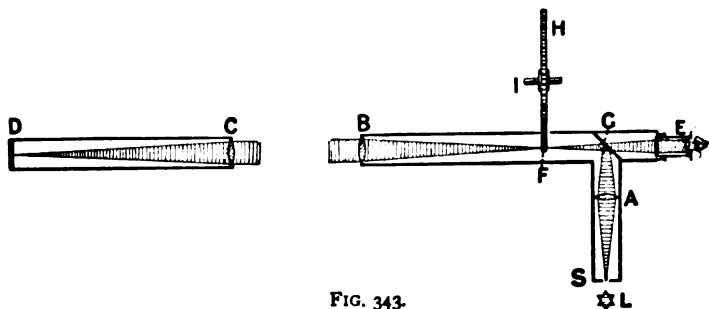


FIG. 343.

star will be seen by the observer, formed by light which has travelled to *D* and back again. A toothed wheel, *H*, which can be rapidly rotated round an axle, *I*, is so arranged that when a tooth passes *F* the light is intercepted, but when a space passes *F* the light is allowed to pass.

If the wheel is slowly rotated, an observer at *E* will see a bright star when a space passes *F*, while when a tooth passes there will be darkness, so that as the wheel rotates the star alternately appears and disappears; but if the speed is such that more than twenty teeth pass per second, owing to the persistence of vision, a permanent star will be seen.

If light took no time to travel from *F* to *D* and back again, then all the light that passed through any space would be able to pass back again through the same space, since by supposition the light takes no time to travel from *F* to *D* and back, and hence the wheel would not have moved between the starting of the light and its arrival back at *F*. If, however, the light takes an appreciable time to travel, then, as the speed of the wheel is gradually increased, it will eventually rotate so fast that by the time the return light reaches *F* the wheel will have turned so that a tooth will have moved round, and will occupy part of the space which was occupied by a space when the light started, so that part of the return light will be cut off by the tooth, and hence the

star seen at E will be of decreased brightness. As the speed is further increased, more and more of the return light will be intercepted by the succeeding tooth, till finally all the light which gets through a space is, on its return, intercepted by the succeeding tooth, and no star is seen. If the speed is yet further increased, the returning light will begin to get through the space next after the one through which it passed on its way out, and a bright star will again be seen. Hence, as the speed is increased, the star alternately appears and vanishes.

If the wheel contains d teeth, then the angle $\angle ACB$ (Fig. 344) subtended by the interval between two consecutive teeth or spaces at the centre will be $360^\circ/d$ or $2\pi/d$ in circular measure. Hence half this angle, or π/d , is the angle through which the wheel has to turn so that a tooth may exactly occupy the position previously occupied by a space.

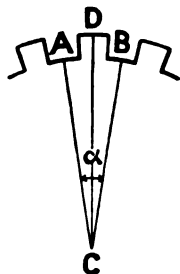


FIG. 344.

If the wheel makes n revolutions per second when the first eclipse occurs, the angle swept out by any radius AC in one second will be $2\pi n$.

Hence the time taken to turn through the angle π/d will be $\frac{\pi/d}{2\pi n}$, or $1/2dn$. If l is the distance between F and D , the distance passed over by the light while the wheel has turned through an angle π/d is $2l$. Hence the light has travelled a distance $2l$ in a time $1/2dn$, or the velocity of light V is given by

$$V = \frac{2l}{\frac{1}{2dn}} = 4ldn.$$

In one of the experiments, l was equal to 8633 metres, the wheel had 720 teeth, and when the star was first eclipsed it made 12.6 revolutions per second, so that

$$V = 4 \times 8633 \times 720 \times 12.6 = 313274000 \text{ metres per second.}$$

More recent experiments made by Cornu, using this method, gave 300,400 kilometres per second, or 186,662 miles per second, as the velocity of light.

367. Foucault's Method of Measuring the Velocity of Light.—In the year 1850, Foucault succeeded in measuring the time light took to travel over a distance of about twenty metres. His method consists in causing a beam of sunlight to fall on a slit S (Fig. 345), by means of a heliostat. The light transmitted by S passes through an unsilvered glass plate G , falls on a convex lens A , and then on a plane mirror B , which can be rapidly rotated round an axis perpendicular to the plane of the figure. For one position of the mirror B , the reflected light falls upon a second mirror C . This latter is a concave mirror, the radius of curvature being equal to $\overline{BC}$. Hence, if the mirror B is at rest, the light reflected at C will retrace its path, being partly reflected at G , so as to form an image of the slit S at D , this image being observed through an eye-piece E . If the mirror B is rotated, the light is only reflected back from C once in each revolution, if the mirror is only silvered on one side. If, further, during the time taken for the light to travel from

B to C and back to B, the mirror has appreciably turned, the return rays will be reflected by B in a slightly different direction to that which they would have taken had the mirror been at rest, and the image D will be displaced to D', the amount of the displacement being read off on a scale placed at D. In order to count the speed of rotation of the mirror B, which was driven by a small steam-turbine, Foucault placed a toothed wheel so that the teeth were illuminated by the intermittent beam of light reflected from the rotating mirror. If the wheel was rotating at such a speed that during the interval between two flashes one tooth had just moved into the position occupied at the previous flash by the pre-

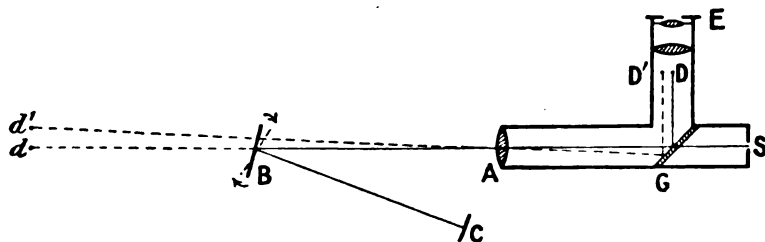


FIG. 345.

ceding tooth, then the wheel would appear to be at rest. When this is the case, the time taken by the mirror to make one revolution is equal to the time taken by the wheel to turn through the angle ACB (Fig. 344), that is, $1/dn$, where d is the number of teeth in the wheel, and n the number of revolutions it makes in a second. Hence the number N of revolutions made by the mirror in one second is dn .

In order to calculate the velocity of light, we require to determine through what angle the mirror B must turn to shift the image formed at the eye-piece from D to D'. Since an image of the slit S is produced at C, we may if we please look upon the problem as to determine through what angle α must B turn so that the image of C formed by the mirror and the lens A may shift from D to D'. Let the two positions of the mirror B be indicated by the full and dotted lines. Then for each position of the mirror B there will be an image of C formed in B. Let d and d' be these images; then, as shown in § 334, d and d' lie on the circumference of a circle having its centre on the axis about which the mirror turns, and passing through C. Further, the arc intercepted between d and d' subtends an angle 2α at the centre of this circle. If, however, the distance BC is called l , we have $\overline{dd'} = 2\alpha l$. Now as far as the lens A is concerned, instead of taking C as the object and considering the two positions of the mirror, we may suppose the mirror B removed and take

the two images d and d' as the objects. Then $\frac{\overline{dd'}}{\overline{DD'}} = \frac{\overline{dA}}{\overline{AG} + \overline{GD}}$. Using

the letters a, b, c to represent the distances $\overline{DD'}$, $\overline{BA}$, $\overline{AG} + \overline{GD}$ respectively, we have

$$\alpha = \frac{\overline{dd'}}{2l} = \frac{a(l+b)}{2lc}.$$

Since the mirror makes N rotations per second, the time it takes to turn through an angle α is $\alpha/2\pi N$, and in this time the light has travelled from B to C and back, that is, a distance $2l$. Hence the velocity, V , of light is given by

$$V = \frac{4\pi Nl}{\alpha} = \frac{8\pi Nl^2 c}{a(l+\delta)}$$

Using this method, Michelson has obtained 299,853 kilometres per second as the velocity of light, with a possible error of ± 60 kilometres.

368. **Aberration.**—A calculation of the velocity of light was made in 1727 by Bradley from the apparent changes which take place in the observed positions of the fixed stars. The simplest method of explaining the principle of this method of calculating the velocity of light is to consider an analogous case. Suppose that a shot is fired from a cannon C (Fig. 346) against a ship AB, which is moving rapidly at right angles to the direction of the trajectory of the shot. If the shot enters the ship at the point D, it will not leave the ship at the point E, for while the shot is travelling across the ship this latter will have moved forward, so that some such point as F will now be in the line CD of the trajectory, and hence the shot will leave the ship at F. Now, as far as the track of the shot left in the ship is concerned, it appears to have come from the point C', or the cannon is apparently displaced in the direction in which the ship is travelling.

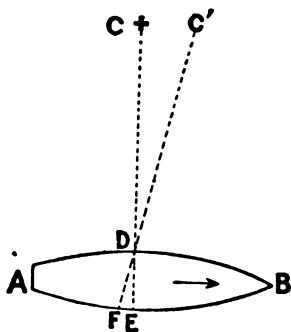


FIG. 346.

Now consider the ship to be replaced by the earth moving in its orbit round the sun, and the shot to be replaced by a light-wave reaching the earth from a star, the line joining the earth and star being at right angles to the direction of motion of the earth. If the earth were at rest, and a telescope D (Fig. 347) were pointed at the star, the axis of the telescope prolonged would pass through the star C. When the earth is moving, however, the telescope is appreciably moved forward in the earth's orbit in the time that elapses between the light-waves reaching the objective D and the eye-piece; and hence, if the axis of the telescope is still directed so as to pass through the star, the waves of light will be left behind and form an image of the star at F to one side of the cross-wires. Just as in the case of the boat, the shot, which if the boat were at rest would have reached the point E, hits the point F. Thus in order to bring the image of the star off to the cross-wires, the axis of the telescope must be pointed along FC'. Now in the time the light has taken to travel from D to F the earth has moved through the distance FE. Hence if the angle CDC', which is called the aberration constant, is α ,

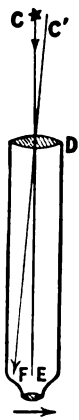


FIG. 347.

and V and v are the velocity of light and of the earth respectively, we have—

$$v/V = \tan \alpha,$$

$$\text{or } V = v/\tan \alpha.$$

Hence if v and α are known, V , that is, the velocity of light, can be calculated.

The above explanation of the phenomenon of aberration is that commonly accepted; it is, however, by no means satisfactory. Thus the quantity V is the velocity of light in the medium, filling the telescope tube, so that if the tube is filled with water, in which the velocity of light is less than in air, we should expect the aberration constant α to increase. No such effect is, however, observed. The discussion of the various theories which have been propounded to explain this discrepancy would lead us beyond the scope of this book, and so we must content ourselves with having drawn attention to the existence of the difficulty.

369. Theories as to the Nature of Light.—We have hitherto assumed that light consists of a wave-motion of some kind, and we now have to consider the evidence on which this assumption is founded.

There have been two principal theories of light. In one of these, which was adopted by Newton, and is called the *emission theory*, a luminous body is supposed to be emitting small particles, called light-corpuscles, which travel out in all directions in straight paths, and all with the same velocity. These light-corpuscles were supposed to cause the sensation of light by their impact on the retina. Since light can traverse not only empty space, but also some forms of matter, these corpuscles must be able to travel through space, and also through matter, which they were supposed to do by passing between the molecules. On this theory the rectilinear propagation of light and the formation of shadows at once followed.

In order to explain the law of refraction on the emission theory, it was assumed that when a corpuscle came near the surface of separation between two media, it was attracted by the denser medium.

Thus let AB (Fig. 348) represent the line of demarcation between two media, say air and water, and IO the path of an incident corpuscle. If v_1 is the velocity with which the corpuscle travels in air, then we may resolve this velocity into

two components, one, $\vec{IN}$, parallel to the surface of separation of the media, and the other, $\vec{IM}$, perpendicular to this surface.

If the water particles exert an attraction on the light-corpuscle when it gets near the surface of separation, this force must act normally to the surface, *i.e.* along ON' . Hence, while the vertical component of the velocity of the corpuscle is increased, the horizontal component, $\vec{IN}$

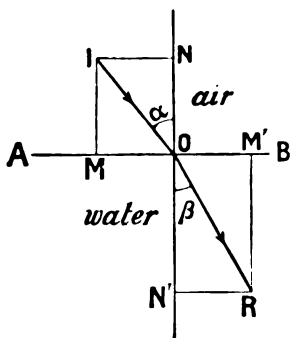


FIG. 348.

will remain unchanged. If then we take $\overline{OM'}$ equal to $\overline{OM}$, and $\overline{ON'}$ equal to the increased vertical component, and complete the parallelogram, $\overrightarrow{OR}$ will represent the velocity of the corpuscle in the water, and its path, that of the refracted ray, will be along OR . From Snell's law we have—

$$a\mu_w = \frac{\sin \alpha}{\sin \beta} = \frac{\overline{IN}}{\overline{IO}} \div \frac{\overline{N'R}}{\overline{RO}} = \frac{\overline{RO}}{\overline{IO}}.$$

Since

$$\overline{IN} = \overline{MO} = \overline{OM'} = \overline{N'R}.$$

But $\overline{IO}$ represents the velocity of the corpuscle in air, while $\overline{RO}$ represents its velocity in water. Hence

$$\frac{\text{Velocity in water}}{\text{Velocity in air}} = a\mu_w,$$

and since the refractive index from air to water is greater than unity, it follows that, according to the emission theory, the velocity of light must be greater in water than in air.

In the second theory, called the undulatory or wave theory of light, a luminous body is supposed to set up vibrations in an all-pervading ether, and these vibrations are supposed to travel through the ether, and when they enter the eye excite the sense of vision. During the passage of the light from the source to the eye, the energy emitted by the source, and which we recognise when it is given up to the retina as light, must be stored up in the ether.

On the older undulatory theory, it was supposed that light-waves consisted of a transverse vibratory movement of the ether itself, but a difficulty was introduced by the fact that, if we suppose that the motion is propagated by the successive parts of the ether setting each other in motion by mutually attracting forces, these forces would be inclined to the direction in which the wave was travelling, and hence they would have a component in the direction of the wave normal, and this component would tend to set up longitudinal waves, in addition to the transverse waves which are required to explain optical phenomena. We have no evidence, however, of the existence of such longitudinal waves in the ether. In the later form of the undulatory theory, called the electromagnetic theory of light, the supposition is made that the vibrations consist not in the change in position of the ether particles, but in a periodic alteration in the electrical and magnetic condition of the ether during the passage of the light. The supposition does not lead to the same difficulty as to the formation of longitudinal waves as does the older theory, and hence possesses a marked advantage.

Since both forms of the theory suppose the existence of a transverse vibration set up in the medium, and only differ as to the nature of the entity the displacement of which constitutes the vibration, the explanations which we shall make in the succeeding sections, since they do not involve the *nature* of the waves, will apply to either form of the theory. We shall also sometimes talk of the displacement of an ether *particle* during the vibration, but this must be taken as a short and convenient

method of stating the displacement of the electric and magnetic condition of the ether at the point under consideration.

We now pass on to consider what assumptions as to the relative velocity of light in air and in water have to be made on the undulatory theory to account for the refraction of light when it passes from air to water. Let AB (Fig. 349) be the line of separation between air and water. Let PP' represent a wave-front in the air, then if v_a is the velocity

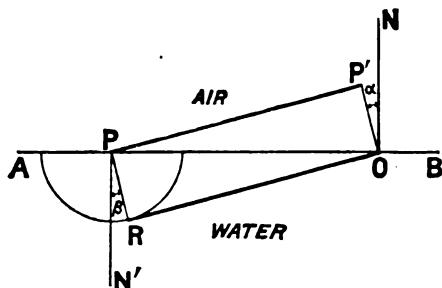


FIG. 349.

of light in air, the time taken for the point P' on the wave-front to reach the second medium will be $P'O/v_a$. During this time the point P on the wave-front will have travelled into the water, and if v_w is the velocity in water, it will have travelled a distance $\frac{P'O}{v_a} \times v_w$. If then, with centre P and radius equal to this distance, we describe a circle, and then from O draw a tangent OR , OR will represent the wave-front in water at the instant when the point P' on the wave-front PP' reaches O (see § 275). If ON and PN' are normals to AB , we have

$$a\mu_w = \frac{\sin NOP'}{\sin N'PR}$$

Now in the triangle $PP'O$ the angle at P' is a right angle, hence the two angles $P'PO$ and $P'OP$ are together equal to a right angle. But the angles $P'ON$ and $P'OP$ are also together equal to a right angle. Hence the angle $P'ON$ is equal to the angle $P'PO$. In the same way, the angle $N'PR$ is equal to the angle POR . So that

$$\begin{aligned} a\mu_w &= \frac{\sin NOP'}{\sin N'PR} = \frac{\sin P'PO}{\sin POR} = \frac{\overline{PO}}{\overline{PO}} \div \frac{\overline{PR}}{\overline{PO}} = \frac{\overline{PO}}{\overline{PR}} \\ &= P'O \div \frac{\overline{P'O}}{v_a} v_w = \frac{v_a}{v_w} \end{aligned}$$

Or the velocity of light in air is to that in water in the ratio of the refractive index from air to water. It will thus be seen that according to the undulatory theory light travels more slowly in water than in air, while according to the emission theory it travels more quickly in water.